

Fully Online Retrieval Adaptation

Executive Summary

21 August 2026

Current evidence

No cluster experiment has yet completed in this new project. Consequently, there is no accuracy conclusion, selected ablation value, or claim that online adaptation improves a frozen backbone.

Implemented protocol

The project now supports a genuinely causal no-split evaluation. Each query retrieves from a balanced complete-date store capped by N_{store} for cross-user retrieval, or up to N_{store} labeled origins for the query user under same-user retrieval, and the ridge or gate fits on exactly N_{fit} already matured rows for the query user. The defaults are 30,000 store windows, 1,000 fitting rows per query user, all-user raw Euclidean retrieval, same-user fitting, and a rolling full ridge. An all-user fitting scope remains available. At every forecast, the oldest 80% of the causal fitting window trains candidate ridges and the newest 20% selects $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and $K \in \{1, 5, 10, 15\}$; the chosen pair is then refitted on the complete window.

Extraction is persistent and independently manifested, allowing changes to adaptation parameters to reuse matching features. Ridge writes rolling coefficient means and standard deviations; CatBoost writes rolling feature importance means and standard deviations. Ridge also records each per-query selected pair and validation loss. Fixed-store/fixed-fit comparisons and every report use a common evaluation-date intersection.

Planned evidence

The first required evidence is the Electricity test profile from `01_main_online_ridge.slurm`. It must establish that checkpoint loading, causal extraction, balanced datastore limits, TS-RAG inference, common-date reporting, and cache size are operational. The second smoke run is `02_online_gates.slurm`, primarily to measure CatBoost refit cost.

Only after these checks should the small profile be run on Electricity, Solar, and Traffic. The full profile then adds raw ETT and prepared TIME datasets. The resulting ablations will characterize alpha/K sensitivity without using final evaluation outcomes for selection, determine whether N_{store} and N_{fit} should differ, and how retrieval and fitting scope interact across users.

Interpretation limit

The Chronos-Bolt SOTA table places causal online MSE beside published TS-RAG and Cross-RAG MSE only as contextual evidence. Their numbers use official test windows and train-split standardization, whereas the new experiment has no fixed split. They must not be interpreted as paired estimates unless a later run reproduces the published evaluation contract exactly.